

Study Report: Build a Large Language Model (From Scratch) - Generative AI Engineer

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Family: Generative AI Engineer
Level: Advanced
Chapters: 7
Source: Reference: Build a Large Language Model (From Scratch) - Sebastian Raschka

Official sources and free libraries

- <https://www.manning.com/books/build-a-large-language-model-from-scratch>

Summary

Study report for **Build a Large Language Model (From Scratch)** by Sebastian Raschka (Advanced level) mapped to the Generative AI Engineer role. Transformer architecture, training, and inference from first principles.

Chapters

Track: Book-Intro

Chapter 1: About Build a Large Language Model (From Scratch) [Advanced]

Chapter focus: About Build a Large Language Model (From Scratch)** *(Level: Advanced)

This study report summarizes **Build a Large Language Model (From Scratch)** by Sebastian Raschka for the Generative AI Engineer role. The resource is rated Advanced level. Transformer architecture, training, and inference from first principles. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Build a Large Language Model (From Scratch)
# Author: Sebastian Raschka
# Role: Generative AI Engineer
# Level: Advanced
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Build a Large Language Model (From Scratch).

When to use it

When learning Generative AI Engineer skills at Advanced level.

Where to use it

Generative AI Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Advanced]

Chapter focus: Why This Book Matters for Your Role** *(Level: Advanced)

Generative AI Engineer professionals use ideas from Build a Large Language Model (From Scratch) to solve real workplace problems. Transformer architecture, training, and inference from first principles. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Generative AI Engineer

Book focus: Transformer architecture, training, and inference from first principles.

Recommended level: Advanced

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Generative AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a advanced skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Advanced]

Chapter focus: Key Topics Covered** *(Level: Advanced)

The main topics in Build a Large Language Model (From Scratch) include practical concepts described as: Transformer architecture, training, and inference from first principles. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Generative AI Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Multimodal Models (Vision + Text) [Advanced]

Chapter focus: Multimodal Models (Vision + Text)** *(Level: Advanced)

GPT-4o, Claude 3.5+, and Gemini accept images + text. Use for diagram explanation, receipt OCR, and UI screenshot debugging. Watch image token pricing.

Code Reference:

```
response = client.chat.completions.create(
    model='gpt-4o',
    messages=[{'role':'user','content':[
        {'type':'text','text':'What error is shown?'},
        {'type':'image_url','image_url':{'url': 'data:image/png;base64,...'}}
    ]}]
)
```

What it shows: image_url content block sends screenshot with question in one call.

Topic guide

What it actually is

Multimodal LLMs reason over pixels and text jointly.

When to use it

Visual troubleshooting, document understanding, and accessibility.

Where to use it

Ed-tech diagram help, insurance claims, and field service apps.

Real use example

Student uploads stats plot - tutor explains axis mislabel from vision input.

Chapter 5: Applied: Production RAG with Re-Ranking [Advanced]

Chapter focus: Production RAG with Re-Ranking** *(Level: Advanced)

Bi-encoder retrieval is fast but coarse; cross-encoder rerankers (Cohere rerank v3, bge-reranker) rescore top-50 to top-5 precisely. Latency tradeoff ~100-200ms.

Code Reference:

```
from cohere import Client
r = cohere_client.rerank(model='rerank-v3.5', query=q, documents=chunks, top_n=5)
```

What it shows: Rerank pushes truly relevant chunk to top for generation.

Topic guide

What it actually is

Re-ranking dramatically improves answer quality on noisy corpora.

When to use it

Large wikis and multi-topic enterprise search.

Where to use it

Compliance corpora where wrong paragraph is costly.

Real use example

Rerank fixes case where BM25 top hit was glossary not procedure.

Chapter 6: Applied: Fine-Tuning vs RAG vs Prompting [Advanced]

Chapter focus: Fine-Tuning vs RAG vs Prompting** *(Level: Advanced)

Prompt when model knows task; RAG when knowledge is private/fresh; fine-tune when style/format/tool-use must be consistent and data is labeled. RLHF/DPO is vendor territory for most teams.

Code Reference:

```
# Decision matrix
# Private docs -> RAG
# Brand tone / JSON tool style -> fine-tune LoRA
# General reasoning -> prompt + best model
```

What it shows: Matrix prevents expensive fine-tune on problems RAG solves.

Topic guide

What it actually is

Architecture choice affects cost, maintenance, and quality ceiling.

When to use it

Greenfield GenAI feature planning.

Where to use it

Support tone fine-tune + policy RAG is common hybrid.

Real use example

LoRA on 2k examples fixes JSON extraction; RAG still supplies facts.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Advanced]

Chapter focus: Study Plan and Practice** *(Level: Advanced)

Finish Build a Large Language Model (From Scratch) with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Generative AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a advanced project screenshot.